

Intrinsic Methods (I): Dunn Index $DI = \frac{\delta}{\Delta}$

- ❑ Intrinsic methods (i.e., no ground truth) examine how compact clusters are and how well clusters are separated, based on similarity/distance measure between objects
- ❑ Dunn Index:
 - ❑ The compactness of clusters: the maximum distance between two points that belong to the same cluster: $\Delta = \max_{C(o_i)=C(o_j)} \{d(o_i, o_j)\}$
 - ❑ The degree of separation among different clusters: the minimum distance between two points that belong to different clusters: $\delta = \min_{C(o_i) \neq C(o_j)} \{d(o_i, o_j)\}$
 - ❑ The Dunn index is simply the ratio: $DI = \frac{\delta}{\Delta}$, the larger the ratio, the farther away the clusters are separated comparing to the compactness of the clusters
- ❑ Dunn index uses the extreme distances to measure the cluster compactness and inter-cluster separation and it can be affected by outliers

Intrinsic Methods (II): Silhouette Coefficient

- Suppose D is partitioned into k clusters: $C_1, \dots, C_k$
- For each object o in D, we calculate
 - $a(o) = \frac{\sum_{o' \in C_i, o \neq o'} \text{dist}(o, o')}{|C_i| - 1}$: avg distance between o and all other objects in the cluster to which o belongs, reflects the compactness of the cluster to which o belongs
 - $b(o) = \min_{C_j: 1 \leq j \leq k, j \neq i} \left\{ \frac{\sum_{o' \in C_j} \text{dist}(o, o')}{|C_j|} \right\}$: minimum avg distance from o to all clusters to which o does not belong, captures the degree to which o is separated from other clusters
- Silhouette Coefficient: $s(o) = \frac{b(o) - a(o)}{\max\{a(o), b(o)\}}$, value range (-1, 1)
 - When the value of o approaches 1, the cluster containing o is compact and o is far away from other clusters, which is the preferable case
 - When the value is negative (i.e., $b(o) < a(o)$), o is closer to the objects in another cluster than to the objects in the same cluster as o: a bad situation to be avoided